

1. print right angle triangle

```
for i in range(1,10):
    print("*" * i)
```

```
*
**
***
****
*****
*****
*****
*****
*****
*****
```

2. Inverted Right angle triangle

```
for i in range(10,1,-1):
    print("*" * i)
```

```
*****
*****
*****
*****
*****
****
***
**
```

3. Right-aligned triangle

```
for i in range(1,10):
    print(" " * (10-i) + "*" * i)
```

```

*
**
***
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*****
*****
*****
*****
```

4. pyramid of stars

```
for i in range(1,6):
    print(" " * (5-i) + "*" * (2*i-1))
```

```

*
***
*****
*****
*****
*****
```

5. Inverted Pyramid

```
for i in range(5,0,-1):
    print(" " * (5-i) + "*" * (2*i-1))
```

```
*****
*****
****
***
**
*
```

6. Number triangle

```
for i in range(1,6):
    for j in range(1,i+1):
        print(j,end=" ")
    print()
```

```
1
1 2
```

◆ What can I help you build?



```

1 2 3
1 2 3 4
1 2 3 4 5

```

7. Inverted number triangle

```

for i in range(5,0,-1):
    for j in range(1,i+1):
        print(j,end=" ")
    print()

```

```

↩ 1 2 3 4 5
   1 2 3 4
    1 2 3
     1 2
      1

```

8. Repeated number of rows

```

for i in range(1,6):
    print(str(i) * i)

```

```

↩ 1
  22
 333
4444
55555

```

9. Continuous count triangle

```

count = 1
for i in range(1,6):
    for j in range(1,i+1):
        print(count,end=" ")
        count += 1
    print()

```

```

↩ 1
  2 3
 4 5 6
 7 8 9 10
11 12 13 14 15

```

10. reverse number triangle

```

for i in range(5, 0, -1):
    for i in range(i, 0, -1):
        print(j, end= " ")
    print()

```

```

↩ 5 5 5 5 5
   5 5 5 5
    5 5 5
     5 5
      5

```

calculator project

```
# Simple Calculator using loops (No functions)
```

```
print("Welcome to the Python Calculator!")
```

```

while True:
    print("\nSelect Operation:")
    print("1. Addition (+)")
    print("2. Subtraction (-)")
    print("3. Multiplication (*)")
    print("4. Division (/)")
    print("5. Exit")

```

```
choice = input("Enter choice (1/2/3/4/5): ")
```

```

if choice == '5':
    print("Exiting calculator. Goodbye!")
    break

```

```
if choice in ['1', '2', '3', '4']:
```

```
try:
    num1 = float(input("Enter first number: "))
    num2 = float(input("Enter second number: "))
except ValueError:
    print("Invalid input! Please enter numbers only.")
    continue

if choice == '1':
    result = num1 + num2
    print(f"Result: {num1} + {num2} = {result}")
elif choice == '2':
    result = num1 - num2
    print(f"Result: {num1} - {num2} = {result}")
elif choice == '3':
    result = num1 * num2
    print(f"Result: {num1} * {num2} = {result}")
elif choice == '4':
    if num2 != 0:
        result = num1 / num2
        print(f"Result: {num1} / {num2} = {result}")
    else:
        print("Error: Cannot divide by zero.")
else:
    print("Invalid choice! Please enter a number from 1 to 5.")
```

 Welcome to the Python Calculator!

```
Select Operation:
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Exit
Enter choice (1/2/3/4/5): 5
Exiting calculator. Goodbye!
```